

Kubernetes QCM – Beginner to Master

BEGINNER LEVEL

1. What is Kubernetes mainly used for?

- A. Writing application code
- B. Managing containers at scale
- C. Designing databases
- D. Monitoring networks

Answer: B

2. What is the smallest deployable unit in Kubernetes?

- A. Container
- B. Node
- C. Pod
- D. Service

Answer: C

3. Which command is used to list all pods?

- A. kubectl get services
- B. kubectl list pods
- C. kubectl get pods
- D. kubectl show pods

Answer: C

4. Which component schedules Pods onto Nodes?

- A. kubelet
- B. kube-proxy
- C. kube-scheduler
- D. etcd

Answer: C

5. What does a Node represent?

- A. A container
- B. A virtual network
- C. A machine running workloads
- D. A storage volume

Answer: C

INTERMEDIATE LEVEL

6. What is the role of a Service in Kubernetes?

- A. Store data
- B. Expose Pods and provide stable networking
- C. Run containers
- D. Schedule Pods

Answer: B

7. Which Service type exposes an application externally on cloud providers?

- A. ClusterIP
- B. NodePort
- C. LoadBalancer
- D. ExternalName

Answer: C

8. What happens if a Pod crashes?

- A. It is permanently deleted
- B. Kubernetes recreates it
- C. Node shuts down
- D. Nothing

Answer: B

9. Which object ensures a number of pod replicas?

- A. Pod
- B. Service
- C. ReplicaSet
- D. ConfigMap

Answer: C

10. What is a Namespace used for?

- A. Network isolation
- B. Resource organization and isolation
- C. Storage management
- D. Pod scheduling

Answer: B

ADVANCED LEVEL

11. What is etcd?

- A. Container runtime
- B. Distributed key-value store for cluster state
- C. Monitoring tool
- D. Load balancer

Answer: B

12. Which object is best for stateless applications?

- A. StatefulSet
- B. DaemonSet
- C. Deployment
- D. Job

Answer: C

13. What is a ConfigMap used for?

- A. Secrets storage
- B. Application configuration
- C. Pod logs
- D. Resource limits

Answer: B

14. What does a DaemonSet ensure?

- A. One Pod per Namespace
- B. One Pod per Node
- C. One Pod per Cluster
- D. One Pod per Service

Answer: B

15. Which command applies changes declaratively?

- A. kubectl create
- B. kubectl replace
- C. kubectl apply
- D. kubectl update

Answer: C

EXPERT / MASTER LEVEL

16. What is the main purpose of RBAC?

- A. Network routing
- B. Authentication and authorization
- C. Pod scheduling
- D. Logging

Answer: B

17. Which component handles networking rules for Services?

- A. kubelet
- B. kube-apiserver
- C. kube-proxy
- D. etcd

Answer: C

18. What is a liveness probe used for?

- A. Traffic routing
- B. Detecting application health and restarting containers
- C. Scaling Pods
- D. Monitoring CPU usage

Answer: B

19. Which resource limits Pod scheduling?

- A. Requests only
- B. Limits only
- C. Requests and Limits
- D. Annotations

Answer: C

20. What is the main difference between StatefulSet and Deployment?

- A. Scaling behavior
- B. Networking
- C. Persistent identity and storage
- D. Image management

Answer: C